

SOFTWARE ENGINEERING LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

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Project: Meal Planning and Diet Management

Jira Board: MPDM Board

1. Objective

The objective of this lab is to create and manage an Agile product backlog using Epics and User Stories, prioritize and estimate the work using Story Points, plan Scrum sprints, simulate sprint progress, and analyze sprint performance using the Burndown Chart.

2. Product Backlog Creation

The project backlog was created in Jira for the Meal Planning and Diet Management system. Four Epics were created to organize the project requirements:

- **Meal Planning and Diet Management**
- **User Profile and Preferences**
- **Meal Subscription Management**
- **Payment and Subscription Billing**

User Stories were created under the respective Epics with appropriate priorities and Fibonacci Story Point estimates.

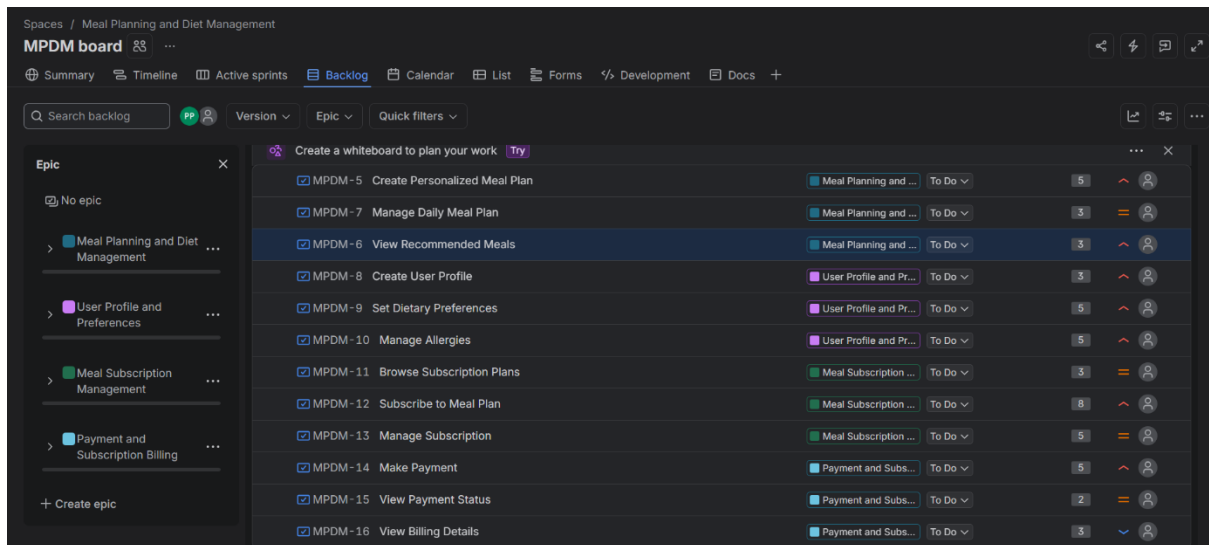


Figure 1: Jira Product Backlog showing Epics, User Stories, priorities and Story Points.

3. Sprint 1 Planning

Sprint 1 was created with a one-week duration. Six User Stories were selected for the sprint with a total estimate of 29 Story Points.

The selected work items were:

Create Personalized Meal Plan – 5 points

View Recommended Meals – 3 points

Create User Profile – 3 points

Set Dietary Preferences – 5 points

Subscribe to Meal Plan – 8 points

Make Payment – 5 points

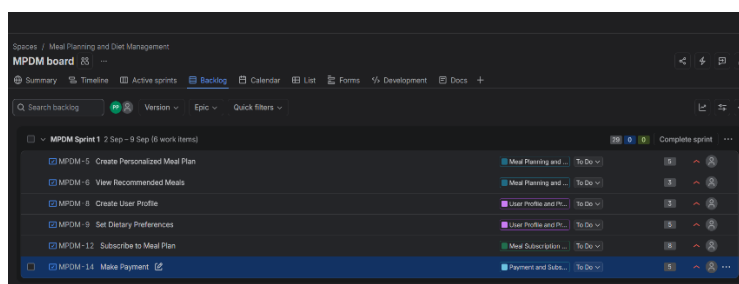


Figure 2: Sprint 1 planning with six selected work items.

4. Sprint 1 Execution

The Sprint 1 work items were simulated through the Scrum workflow by moving them from To Do → In Progress → Done.

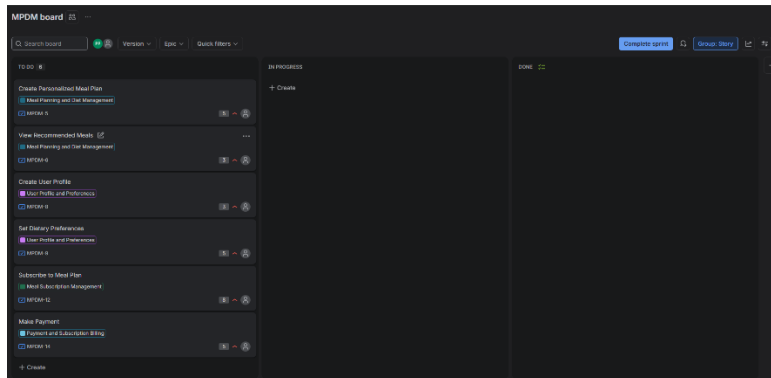


Figure 3: Sprint 1 board showing the TODO of User Stories.

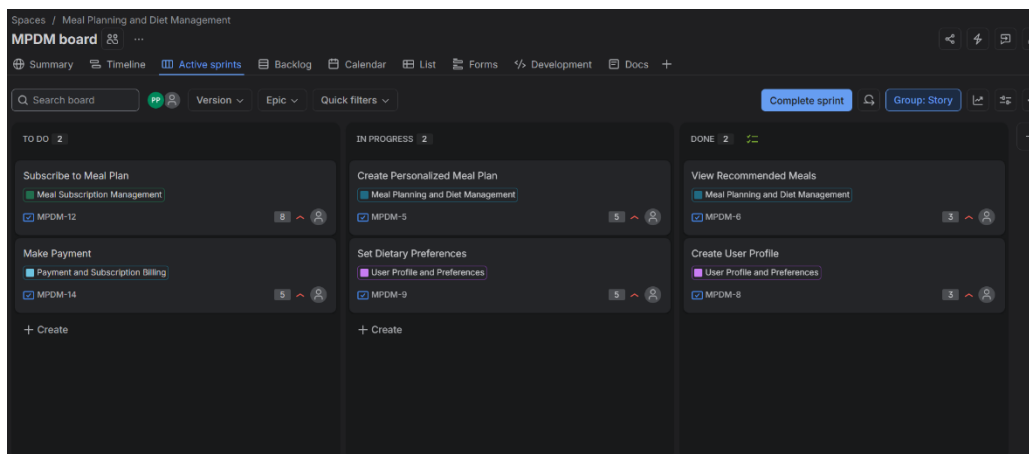


Figure 4: Sprint 1 Few User Stories moved to IN PROGRESS And DONE

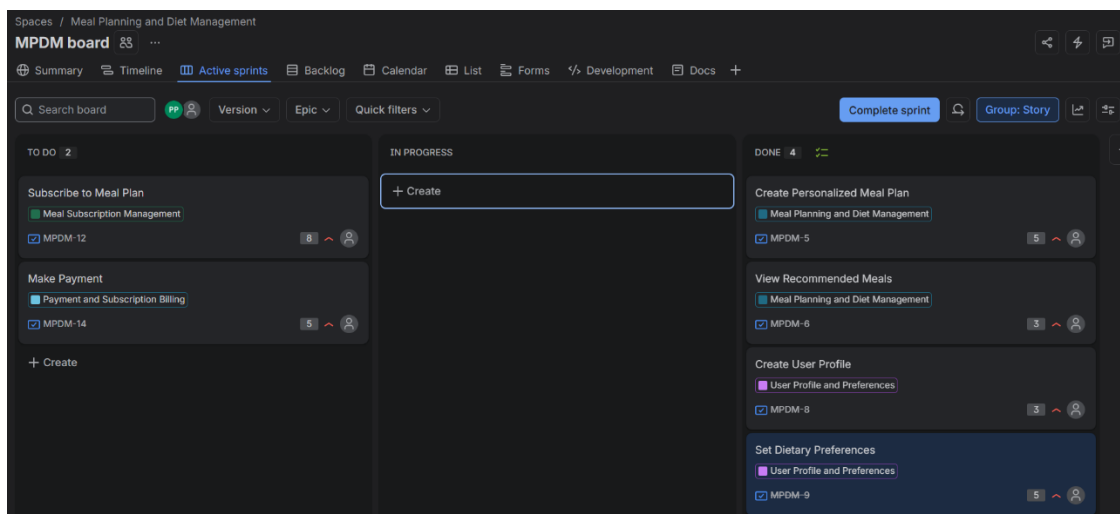


Figure 5: Sprint 1 Few User Stories moved to IN PROGRESS And DONE

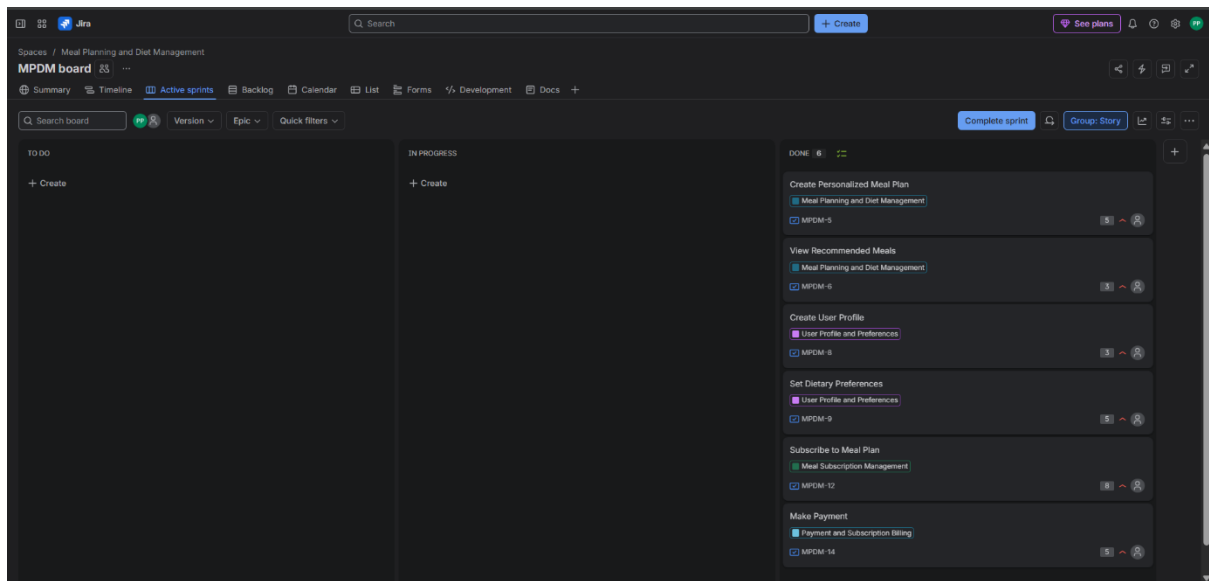
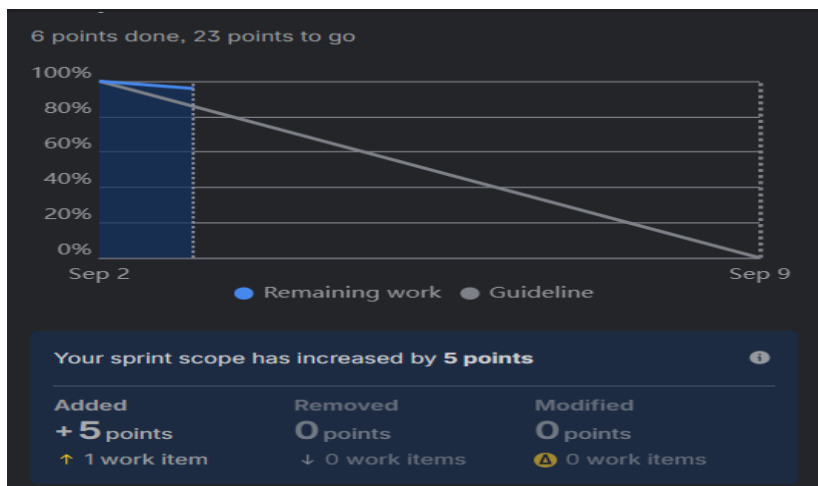


Figure 6: Sprint 1 completed with all selected User Stories moved to Done.

5. Sprint 1 Burndown Analysis

The Sprint 1 Burndown Chart was used to monitor the remaining work during the sprint. The chart showed the completed and remaining Story Points and also reflected the change in sprint scope.



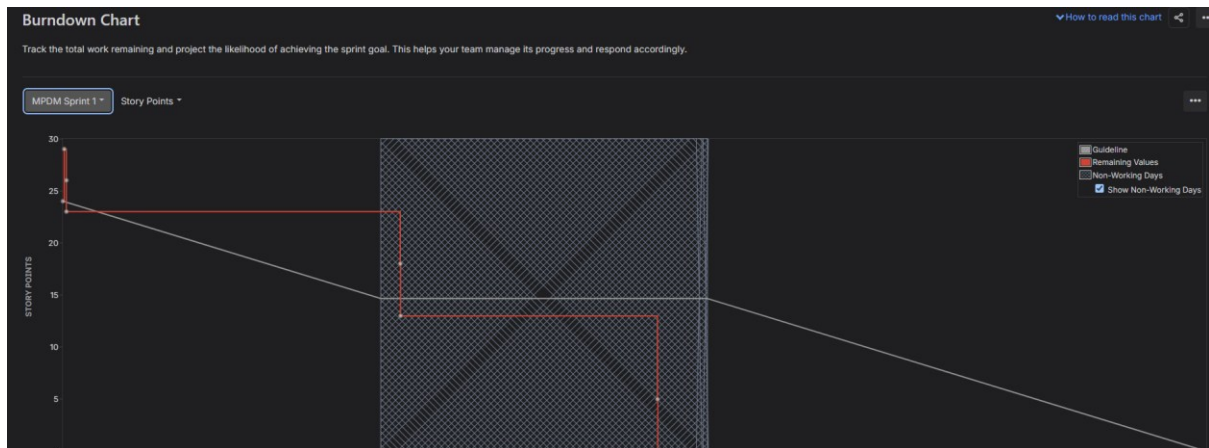


Figure 7: Sprint 1 Burndown Chart showing sprint progress and remaining work.

6. Sprint 2 Planning

After completing Sprint 1, Sprint 2 was planned with the remaining User Stories.

Sprint 2 contained 6 work items with a total estimate of 21 Story Points.

The selected stories were:

Manage Daily Meal Plan – 3 points

Manage Allergies – 5 points

Browse Subscription Plans – 3 points

Subscription – 5 points

View Payment Status – 2 points

View Billing Details – 3 points

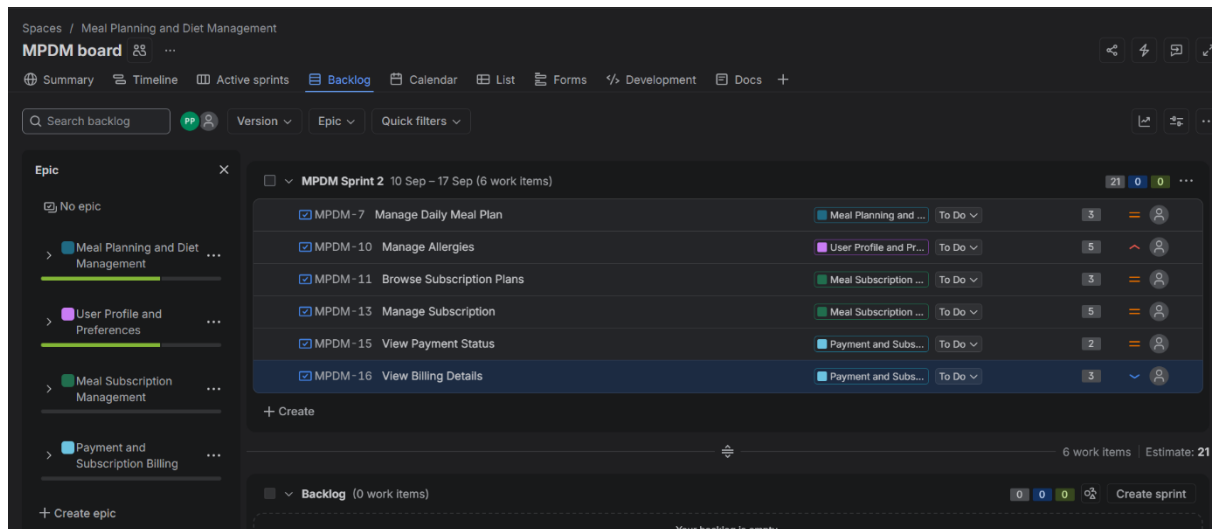


Figure 8: Sprint 2 planning showing six work items and 21 Story Points.

7. Sprint 2 Execution

The Sprint 2 User Stories were simulated using the Scrum workflow. Each story was moved from To Do → In Progress → Done.

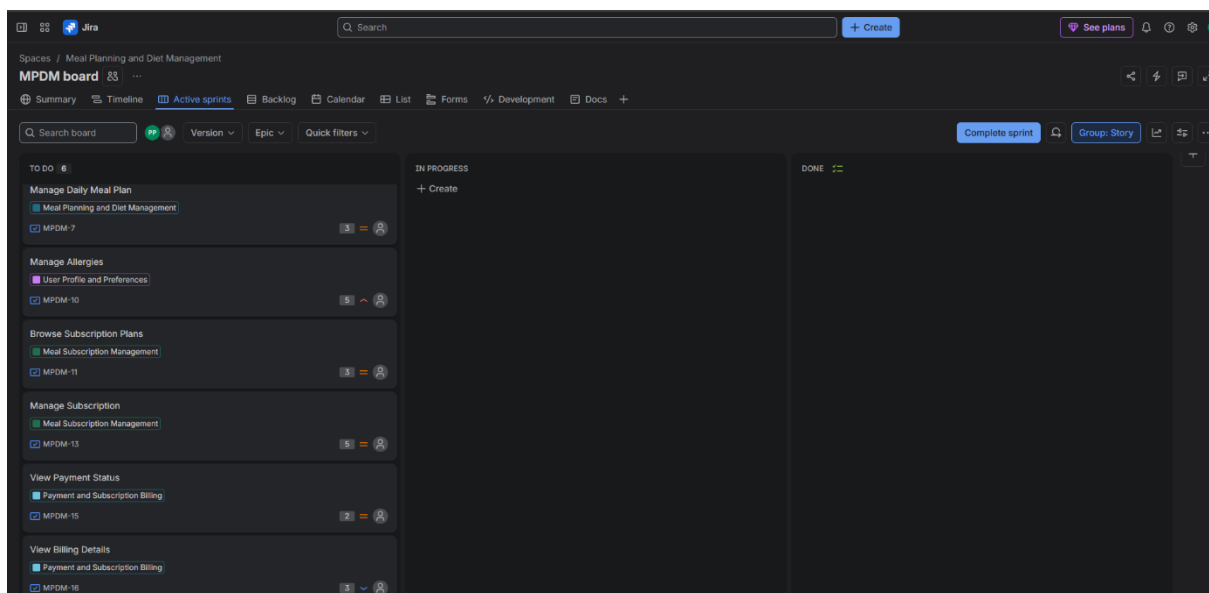


Figure 9: Sprint 2 active board at the beginning of the sprint.

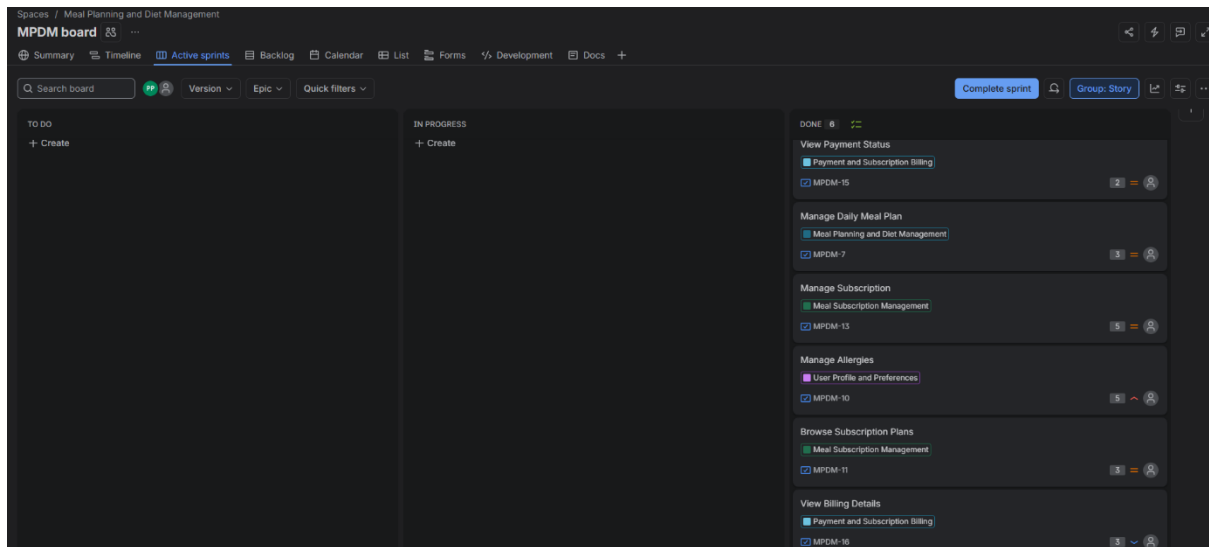


Figure 10: Sprint 2 board showing all six User Stories completed.

8. Sprint 2 Burndown Chart

The Sprint 2 Burndown Chart shows that all 21 Story Points were completed, resulting in 0 points remaining.

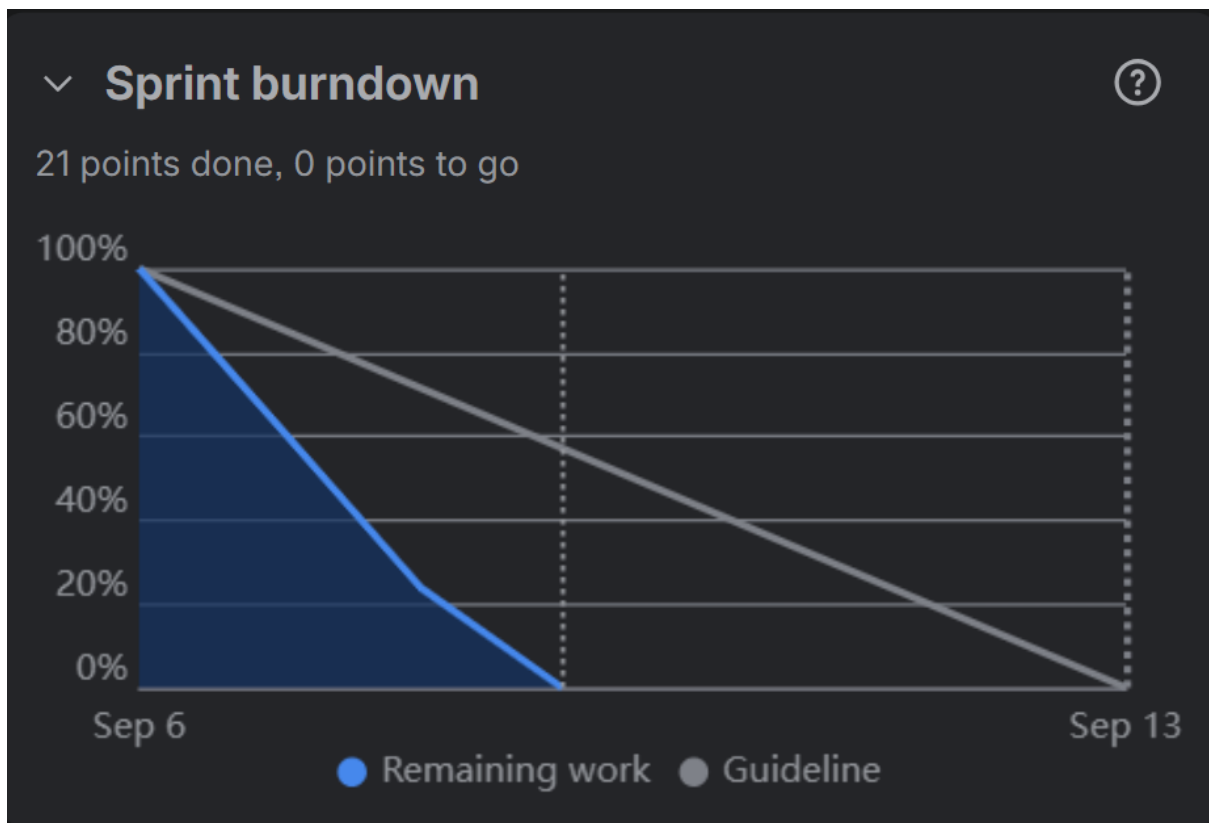




Figure 11: Sprint 2 Burndown Chart showing 21 points completed and 0 points remaining.

9. Reflection

1. Did your estimations reflect the actual effort?

The Fibonacci Story Point estimates provided a useful way to represent the relative effort and complexity of each User Story. The estimates were suitable for the simulated sprint planning and helped in deciding how much work could be included in each sprint.

2. Was your backlog well-prioritized?

Yes. Important features such as creating meal plans, managing user profiles, subscribing to meal plans, and making payments were given higher priority. Supporting features were assigned medium or lower priority based on their importance.

3. How did your simulated sprint align with your plan?

The User Stories were first organized in the product backlog and then divided into two sprints. During execution, the stories were moved through To Do, In Progress, and Done, which demonstrated the Scrum workflow in Jira.

4. What insights did the Burndown Chart give about your team's capacity?

The Burndown Chart showed the progress of the sprint and the amount of work remaining. In Sprint 2, all 21 committed Story Points were completed, resulting in zero remaining work. This provides useful information for estimating the team's capacity when planning future sprints.

10. Conclusion

This lab demonstrated the basic Agile and Scrum workflow using Jira. The activities included creating Epics and User Stories, prioritizing requirements, assigning Story Points, planning two sprints, simulating progress, completing the sprints, and analyzing sprint progress using the Burndown Chart.